Macaca mulatta (rhemac10) snRNAseq analysis

Rhemac10 snRNAseq Pipeline Overview

1. Ran Cellranger Count with BaDoi's original rhemac10 lifoff annotation (rheMac10_liftoff_GRCh38.p13_RefSeq.gtf) and UCSC FASTA
2. Perform downstream analysis in R (some functions redundant with Cellranger)
   - SoupX - Identify ambient RNA contamination
   - MiQC – Identify damaged nuclei
   - DropletQC – Identify empty droplets
   - Seurat – Analyze snRNAseq data
   - DoubletFinder – Identify doublets
3. Combine all libraries into one R object to create combined UMAP
4. Create feature plots
5. Finalize cell types in each cluster
6. Discover cell states (active vs inactive)
7. Compare to other macaque snRNAseq datasets

Performed only on 7 libraries ~56k nuclei

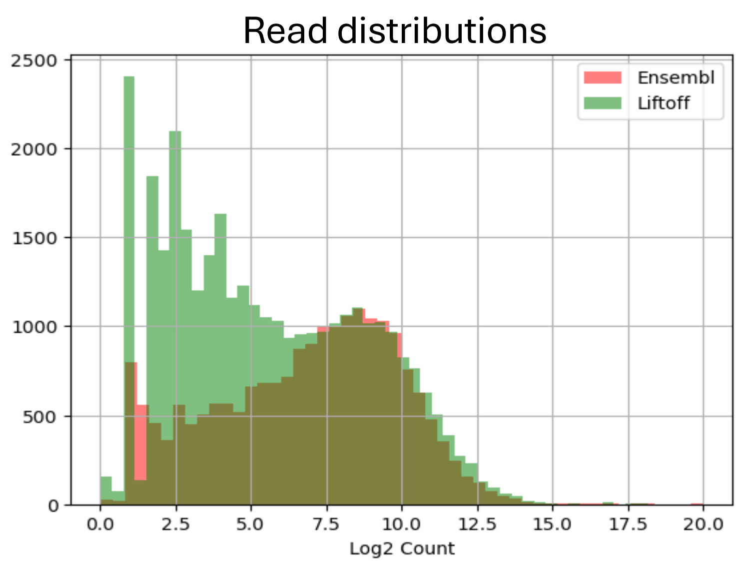
All libraries sequenced & in cluster

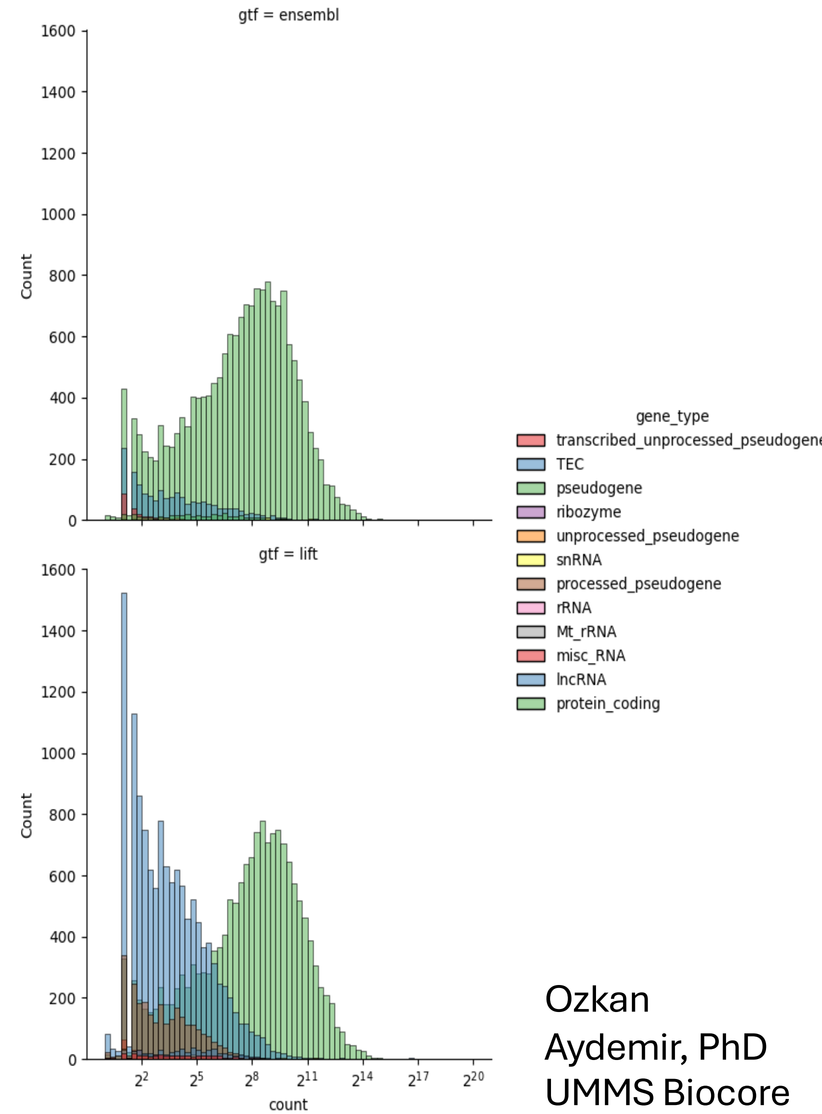
- 18 libraries/technical replicates sequenced
  - 8 reps from MI1
  - 2 reps from MI2
  - 8 reps from MI3
- Conditions
  - MI1- monocular inactivation animal 1
  - MI2- monocular inactivation animal 2
  - MI3 - monocular inactivation animal 3
- Combined gene expression (GEX) library 3 fastq files from Run6 and Run9

experiment	sequencing Run	Sample ID*
MI2	1	3-C_MI2_230608
MI2	1	4-D_MI2_230608
MI1	1	1-MI1_230808
MI1	1	2-MI1_230808
MI1	1	4-MI1_230808
MI1	1	5-MI1_230808
MI1	1	7-MI1_230808
MI1	3	3-MI1_V1_230809
MI1	3	6-MI1_V1_230809
MI1	3	8-MI1_V1_230809
MI3	3	1-MI3_V1_231103
MI3	3	2-MI3_V1_231103
MI3	3	3-MI3_V1_231103
MI3	3	4-MI3_V1_231103
MI3	6 & 9	5-MI3_V1_231103
MI3	6 & 9	6-MI3_V1_231103
MI3	6 & 9	7-MI3_V1_231103
MI3	6 & 9	8-MI3_V1_231103

Using Ensembl genome files for mapping

- Similar read totals mapping with two annotations but Liftoff Rhemac10 had an increase in lower to mid range counts whereas Ensembl Mmul_10 had a few very highly expressed genes
- Read distribution of ensembl annotation for Mmul_10 closer to normal distribution
- Ensembl annotation contains more protein coding genes whereas liftoff contains more lncRNA's
- Will move forward with ensembl for snRNAseq & multiomics (macfas6)





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Experimental Plan

1. Obtain approx. 200k high-quality nuclei from all libraries sequenced
2. Finalize clustering and ID cell types
   → Integrate dataset with Wei et al. (2022). Identification of visual cortex cell types and species differences using single-cell RNA sequencing. *Nature Communications,* 13, Article 6902. https://doi.org/10.1038/s41467-022-34590-1
3. Identify activated vs. inactivated transcriptional profile for each neuronal cell type
   → Predict "active" cells based on expression of small number of known activity-regulated genes in:
   - All Excitatory neurons
   - Layer IV excitatory neurons
   - All Inhibitory Neurons
   - VIP neurons (doi: 10.1038/nature17187)
   - Any subtypes for which we can reliably call "active" vs "inactive" nuclei

   → Validate using RNA FISH in tissue sections
4. Identify novel neuronal activity-dependent gene expression patterns by comparing Macaque activity-dependent transcriptomes with those from Mouse V1 scRNAseq (DOI: 10.1038/s41593-017-0029-5)

Next steps

- Re-map all snRNAseq data together using ENSEMBL Rhemac10 GTF (no chr) & FASTA (unmasked) with 10x cellranger count
  - GTF: Macaca_mulatta.Mmul_10.113.gtf
  - FASTA: Macaca_mulatta.Mmul_10.dna.toplevel.fa.gz
- Perform downstream analysis with Seurat R
  - Will have a lot more cells across libraries
  - Removing SoupX & DropletQC for initial QC (redundant)
  - Will keep current cutoffs (e.g. UMI > 1000)
  - Initial UMAPS should be slightly higher resolution than before (e.g. need to separate LAMP5 GABAergic neurons from VIP neurons)
  - Once we feel good about labelling/identifying clusters, we can provide snRNAseq UMAP and metadata to Via Scientific with Multiomics mapping output and request their help with analysis.